Mem-X

*Can embodied agents **recall past experience**, **localize across episodes**, and **follow historical routes** when **memory** is required?*

L1 Memory Recall



Shared global coordinate frame



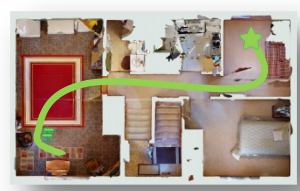
Recall a past target

"Walk to the sink you saw earlier."

L2 In-Memory Localization



Start within a previously visited region



Localize + Index

"Go to the bed seen earlier, then stop at the curtains."

L3 Off-Memory Relocalization



Start outside historical memory



Relocalize + Retrace

"Return to the dining table, then retrace the route to the bathtub."

(a) Three Levels of Memory-Dependent Cross-Episode Navigation

Prior Benchmarks

C: Cross-episode memory
M: Memory-required tasks
L: Self-localization

Benchmark	C	M	L
R2R	✗	✗	✗
IVLN	✓	✗	✗
FindingDory	✓	✓	✗
Mem-X	✓	✓	✓

Explicit memory dependencies

Mem-X

L1 Recall
Single-trajectory memory

↓

L2 Localize + Index
Find the correct episode

↓

L3 Route Re-entry
Align a historical sub-path

MemBridge

RGB-D + local pose

Scene-graph construction

↓

Cross-episode localization

↓

Memory alignment + fusion

↓

Coarse-to-fine retrieval

Evaluation

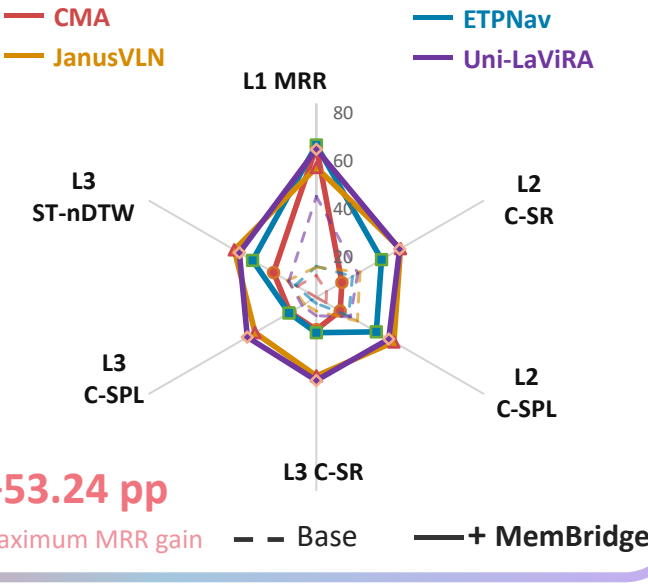
Existing VLN agent
(CMA, ETPNav, JanusVLN, Uni-LaViRA)

↓

Memory-aware metrics
MRR Loc-SR
C-SR C-SPL
ST-nDTW

(b) Progressive Benchmark and Memory Evaluation

Memory-task metrics (%)



(c) Performance Comparison